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Interim Assessment Report

Boundary Evaluation of Low-Fidelity sEMG Acquisition for Dexterous Hand Control

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Agenda

1. Background & Objectives

Exploring sEMG (Surface Electromyography) as a Human-Robot Interface (HRI) under hardware constraints and defining our boundary evaluation approach

2. System Architecture & Methodology

The control system blueprint: from signal acquisition to tendon-driven actuation

3. Current Progress

Hardware assembly, signal pre-processing, and vision-based tracking setup

4. Future Plan & Timeline

Algorithm evaluation roadmap and deployment Gantt Chart

1. Background & Objectives

- **sEMG as a Human-Robot Interface(HRI)**

Surface Electromyography offers an intuitive pathway for translating human motor intent into teleoperation and robotic actuation

- **The Hardware bottleneck (Repurposed Sensors)**

Utilising low-cost biopotential modules (e.g., AD8232 ECG front-end) for sEMG acquisition results in severe high-frequency information loss (>40Hz cutoff). This makes complex continuous control fundamentally unstable

- **Core Objective**

To systematically evaluate the **algorithmic and physical boundaries** of utilizing low-fidelity sEMG acquisition for tendon-driven hand control

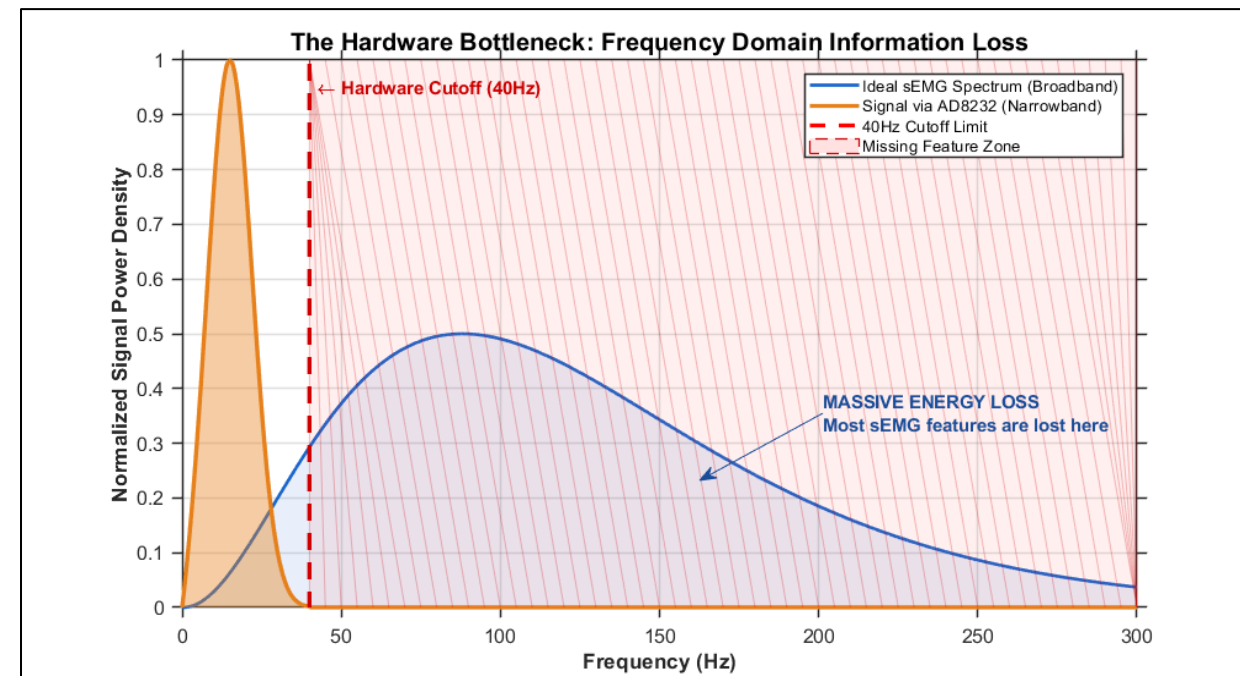
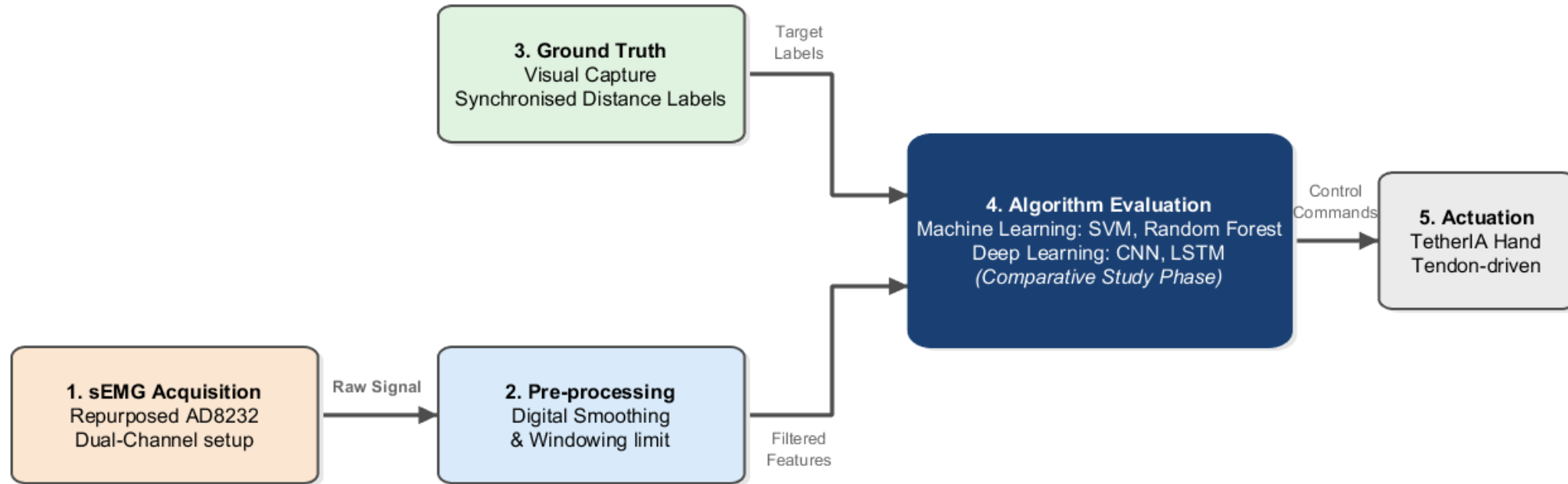


Fig. 1, The Hardware Bottleneck. Spectral truncation of sEMG signals at 40Hz, defining the physical boundary for algorithmic compensation and control evaluation

2. System Architecture & Methodology



Dual-Track Pipeline

Contrasting high-fidelity visual ground truth against low-fidelity physical sEMG inputs to create a rigorous testing environment

Tiered Algorithm Evaluation

Deploying classic (Random Forest) vs. SOTA (LSTM) models to map the exact degradation limits caused by hardware bottlenecks

Hardware-in-the-Loop

Evaluating the control pipeline directly on the assembled TetherIA hand, prioritising physical actuation constraints

2.1 Progressive Algorithm Evaluation Strategy



Level 1: Baseline Binary Classification	Level 2: Advanced State Classification	Level 3: Temporal Sequence Analysis	Level 4: Complex Intent Recognition
Support Vector Machine (SVM)	Random Forest	Long Short-Term Memory (LSTM) network	CNN
Basic Binary recognition (rest vs action)	Complex binary or basic multi-class recognition with increased gesture variations	Continuous gesture recognition leveraging long-term temporal dependency analysis	Fine-grained muscle intent recognition and force/angle prediction for natural and precise control

Table 1: Four-tier algorithm evaluation framework. The strategy demonstrates the progression from basic binary rest-action classification using traditional algorithms to fine-grained intent and force prediction using advanced neural networks.

3. Current Progress

3.1 Hardware Integration & Actuation Readiness



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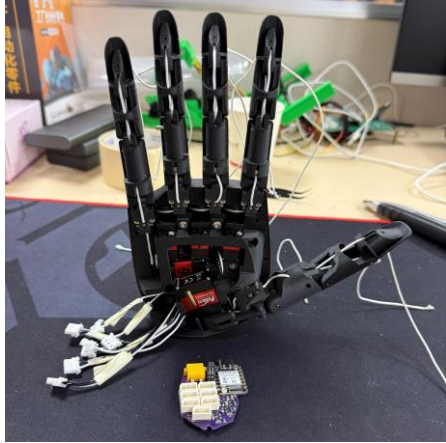


Fig. 2, Physical assembly and bench testing of the TetherIA robotic hand.

Hardware Platform Adoption

Prioritise the use of the verified open-source TetherIA platform to achieve system-level control evaluation, rather than redundant mechanical design

Bio-inspired Tendon-driven Mechanism

The TetherIA hand utilises a tendon-driven system to mimic human hand anatomy. By locating actuators in the palm and using cables to transmit force, the design achieves a high degree of dexterity while maintaining low finger mass

Physical Assembly & Debugging

Completed the physical construction, debugging the servo actuation and tethers to ensure mechanical reliability for practical experiments

3.2 Acquisition Hardware & Signal Setup

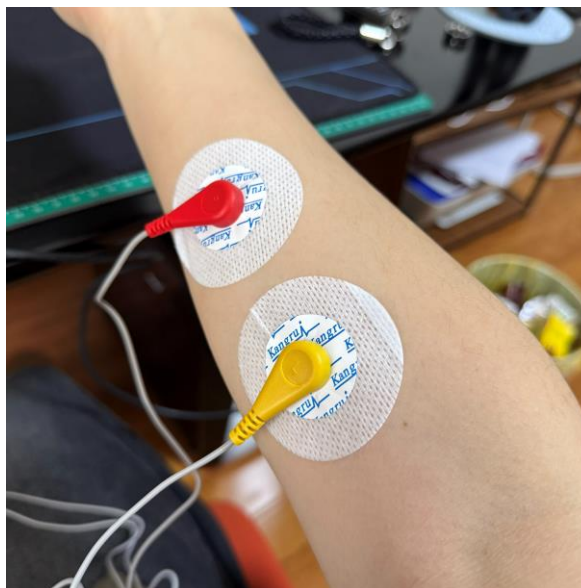
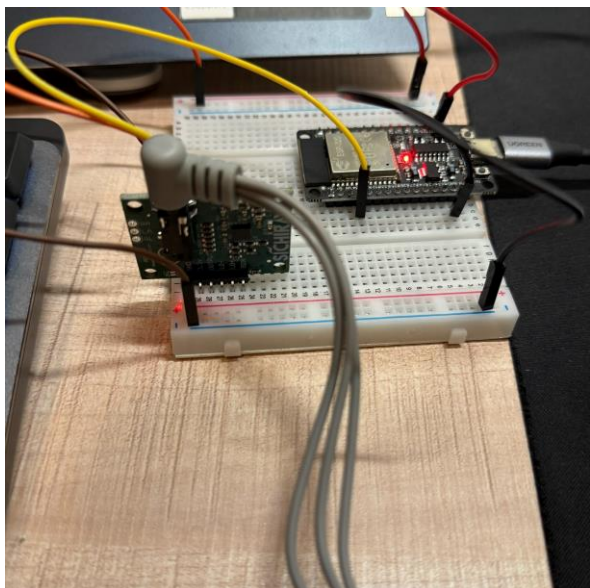


Fig 3, Prototype circuit on ESP32 and electrode placement for sEMG acquisition

Circuit Prototype: Built an acquisition circuit using ESP32 and AD8232 modulus on a breadboard

Electrode Placement: Standardised the electrode positioning on the forearm to ensure consistent signal quality for muscle contraction

Analogue Processing: Initial testing confirms successful conversion of muscle activity into digitised data

3.3 Control Logic & Real-time verification

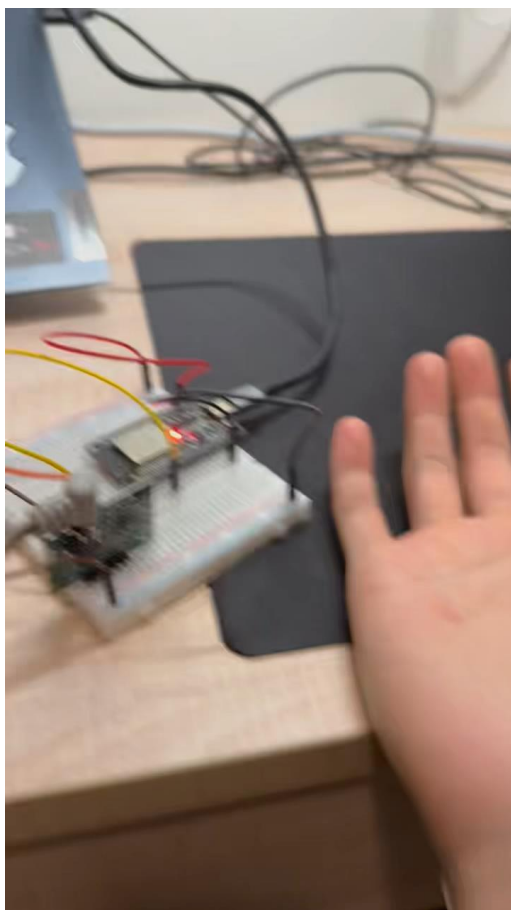
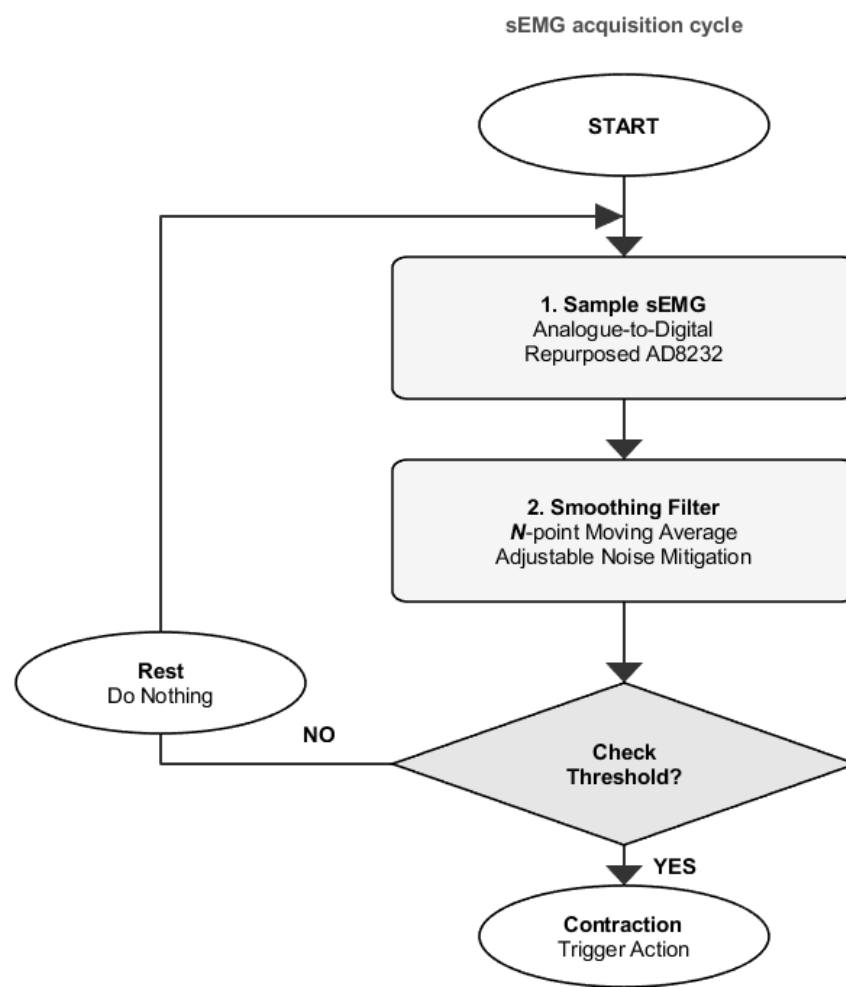


Fig. 4, Real-time sEMG signal acquisition and threshold testing



Initial Calibration: The system establishes a baseline and an active threshold based on user relaxation and maximum contraction states

Signal Processing: Analogue sEMG signals are digitised and smoothed using an N -point moving average filter before threshold comparison

Preliminary Verification: The current setup uses binary thresholding to map contractions to simple actions. This verifies the sensor and processing pipeline, serving as a baseline for the upcoming multi-level control implementation

3.4 Vision-Assisted Data Labelling

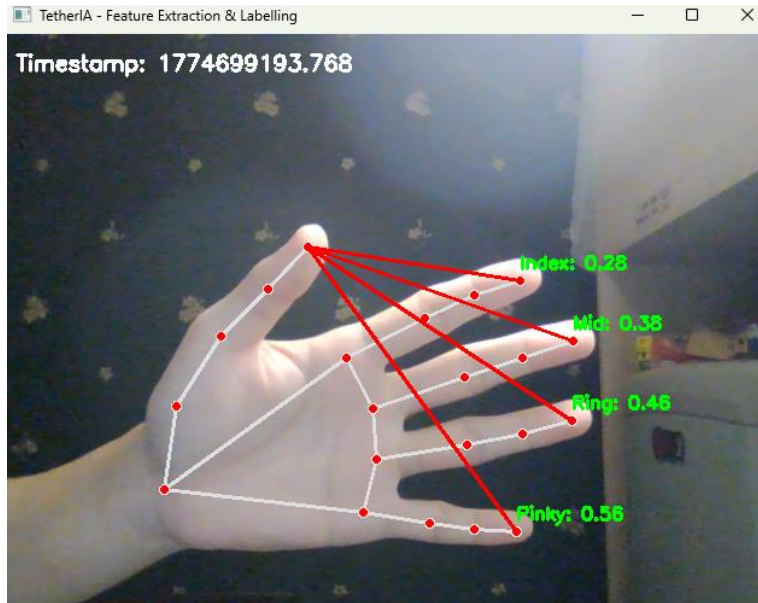
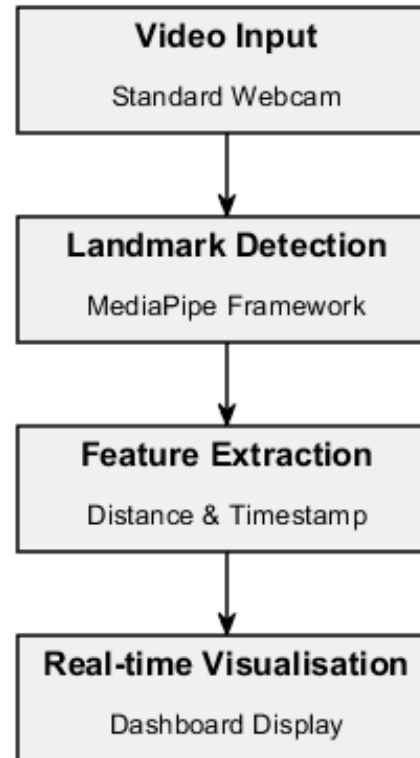


Fig. 5, Real-Time hand mark tracking and distance calculation



Ground-truth Generation: Utilising a standard webcam to automate sEMG data tagging

Kinematic Tracking: Real-Time extraction of hand landmarks via the MediaPipe framework

Feature Selection: Quantifying hand closure through thumb-to-fingertip distances

Data Alignment: Integrating high-precision timestamps for future sEMG synchronisation



4. Future Plan & Timeline

Work Package	Task ID	Task Description	Prerequisites
WP1: Data Pipeline & Hardware Setup	T1	Procurement and integration of 2nd sEMG Module	None
	T2	Hardware Debugging & Signal Pre-processing Optimisation	T1
	T3	Development of Automated Data Annotation Pipeline via visual synchronisation	T2
WP2: Progressive Algorithm Evaluation	T4	Level 1& 2 tuning: Baseline binary and advanced state classification (SVM, Random Forest)	T3
	T5	Level 3 tuning: Temporal sequence analysis for continuous gesture recognition (LSTM)	T4
	T6	Level 4 tuning: Complex intent recognition and fine-grained prediction (CNN)	T5

Work Package	Task ID	Task Description	Prerequisites
WP3: System Validation & Deployment	T7	Kinematic simulation validation (Sim-to-Sim)	T6
	T8	Physical deployment and tendon-driven actuation testing (Sim-to-Real)	T7
WP4: Thesis Writing	T9	Outline finalisation, Introduction and literature review	None
	T10	Methodology drafting	T3
	T11	Result analysis, Discussion and Future work	T8
	T12	Final report content and formatting check	T11

4. Future Plan & Timeline

